

## AUGUSTINE ATTA DEBRAH

Analytical Chemistry • Method Development • Instrumentation • Separations & Mass Spectrometry  
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### PROFILE

Analytical chemist completing a Ph.D., with 11 peer-reviewed publications in chromatography, mass spectrometry, and chemometrics - including method development for pharmaceutical extractables and non-targeted contaminant screening. Builds analytical instrumentation end-to-end and develops validated quantitative methods across GC, LC, MS, and spectroscopic platforms. Seeking an analytical development, R&D, or applications scientist role.

### TECHNICAL SKILLS

**Separations & MS:** HPLC/UPLC, GC-FID/ECD, GC-MS, UPLC-QTOF-MS, UPLC-MS/MS, ion mobility-MS, supercritical fluid chromatography, HILIC, CE-LIF, SPME, flow injection analysis

**Spectroscopy & Wet Chemistry:** Raman, FTIR, UV-Vis, LIBS, AAS; redox titrimetry, acid digestion, spectrophotometric determination

**Methods & Quality:** Method development and validation, calibration, sample preparation and extraction, quantitative and non-targeted analysis, chemometrics, process monitoring and optimization, SOP authoring, QC documentation, instrument prototyping and troubleshooting, preventive maintenance

**Data & Software:** Python (data and image analysis), JMP (DOE / statistical modeling), MZmine, MaxQuant, ChemDraw, Multisim, BioRender, MS Office

### RESEARCH & TECHNICAL EXPERIENCE

**Graduate Research Assistant** - Stockton Laboratory, Georgia Institute of Technology **2022 - Present**

- Designed, built, and validated an aqueous-in-gas microfluidic droplet system from concept through commissioning and performance qualification; established stability regimes and characterized flow-rate-dependent size distributions and operating transitions.
- Developed CE-LIF methods for quantifying fluorescently labeled amino acids, establishing concentration stability, precision, and analyte recovery across droplet formation and collection steps.
- Built Python data and image-analysis pipelines for high-speed droplet imaging, replacing manual measurement and accelerating design iteration.
- Conduct spectroscopic and optical characterization of droplet systems; author protocols and diagnose device failures. Funded by Scialog (Research Corporation for Science Advancement).
- Mentor four undergraduate researchers on instrument development, experimental design, data analysis, and laboratory safety.

**Graduate Teaching Assistant, Analytical Instrumentation** - Georgia Institute of Technology **2022 - Present**

- Operate and instruct on GC, GC-MS, LC, Raman, FTIR, UV-Vis, LIBS, and flow injection analysis; trained 100+ students in sample preparation, method development, calibration, and data interpretation.
- Author and revise Standard Operating Procedures to ensure consistent, traceable laboratory practice; provide real-time instrument and method troubleshooting.

**Laboratory Technologist & Research Assistant** - University of Cape Coast, Ghana **2018 - 2022**

- Designed and tested custom circuit systems and laboratory instrumentation; implemented preventive maintenance and calibration schedules sustaining instrument uptime and safety.
- Instructor of record for instrumentation and electronics laboratories (100+ students/year); supervised undergraduate research in electronics, microbiology, and food science.

**Laboratory Assistant (Intern)** - Ghana Manganese Company Ltd., Ghana **2015**

- Performed manganese determination by redox titrimetry following acid digestion, and boron analysis by spectrophotometry on water and environmental samples; prepared ore samples and standard solutions.
- Documented analytical results and supported quality control procedures across the assay and environmental laboratories.

### SELECTED PUBLICATIONS (11 TOTAL)

Sun, X. et al. incl. **Debrah, A. A.** (2021). Supercritical fluid extraction with UPLC-QTOF-MS for determination of extractables to evaluate drug-rubber closure compatibility. *AAPS PharmSciTech*, 22.

Wang, M. et al. incl. **Debrah, A. A.** (2023). Selective extraction of phospholipids using glass fabric modified with a zirconium-based metal-organic framework. *J. Chromatography A*, 1710, 464435.

Ma, X. et al. incl. **Debrah, A. A.** (2021). Risk classification strategy for food-contact material migrants combined with three in silico (Q)SAR tools. *J. Hazardous Materials*, 419, 126422.

### EDUCATION

**Ph.D., Analytical Chemistry** - Georgia Institute of Technology (Advisor: Prof. Amanda Stockton) (Expected 2027)

**Certificate**, Technology Innovation, TI: GER Program, Scheller College of Business (Expected Dec. 2026)

**M.S., Chemistry** (2025)

**B.Sc., Laboratory Technology** - University of Cape Coast, Ghana (2016)

### AWARDS, LEADERSHIP & AFFILIATIONS

**Awards:** White & Gold Student Leadership Award, Georgia Tech (2024) • AbGradCon Conference Award (2025) • GT College of Sciences Travel Grant (2026)

**Leadership:** Founder & Program Lead, Analytical Instrumentation Training Program, Georgia Tech (2025-Present) • Graduate Representative, UGAP Academic Programs Committee • President, African Graduate Students Connect

**Conferences:** ACS Spring 2026 • Pittcon 2025 • ASME FEDSM 2025 (oral) • Member: ACS, AOAC, SLAS, SETAC